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Deformation and Energy Absorption of meta-Structure-reinforced Granular Composites under Triaxial loading

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Deformation and Energy Absorption of Metastructure-reinforced Granular Composites under Triaxial loading

Abstract: Metastructure has been proved to possess distinct energy-absorbing capability under uniaxial loading condition. However, its performance under complex loading conditions was rarely reached. This study attempts to evaluate the deformation and energy-absorbing characteristics of a metastructure-embedded granular composite. Fabricated via 3D printing for precise auxetic structural control, the metastructure exhibits distinct negative Poisson's ratio effect. A series of laboratory tests using monotonic, constant shear stress (CSS), and cyclic triaxial stress paths were carried out, to investigate the mechanical response of the composites. Results demonstrate that the inclusion of the metastructure significantly enhance the composite strength and instability resistance. Triple-layer reinforcement increased peak and residual strengths of the composite by approximately 40% while delaying peak stress mobilization. During cyclic loading, energy absorption exhibited a non-linear enhancement, increasing by roughly 90% and 200% for single and triple-layer configurations, respectively. To elucidate this non-linearity, a mass-normalized specific net energy index was introduced to quantify the effect of overlapping confinement zones on per-unit-mass efficiency. Meanwhile, the damping ratio slightly decreased with metastucture inclusion. These findings highlight the exceptional value of metastructure in optimizing energy absorption under complex stress conditions.

Keywords: Auxetic; Metastructure; Energy absorption; Triaxial test; Instability.

1. Introduction

Mechanical metastructures can be engineered through geometric and topological design to achieve tailored mechanical responses [20, 41, 53]. This design flexibility has enabled promising applications in aerospace, transportation, protective structures, and precision engineering. Representative examples include energy-absorbing systems for spacecraft soft landing, crashworthiness enhancement, and impact mitigation [7, 10, 28]. These structures are also utilized for precision instruments, providing platforms for low-frequency vibration isolation, wave filtering, and vibration shielding [1, 30, 32]. The distinctive mechanical performance of these metastructures originates primarily from their engineered microstructural structures, rather than solely from the intrinsic properties of the constituent material. Unlike conventional homogeneous materials, their macroscopic mechanical responses and effective constitutive behaviors are largely governed by the geometry of the unit cell and the way unit cells are arranged in space [21, 34]. By tailoring cell morphology, connectivity, and spatial arrangement, mechanical metastructures can exhibit unconventional mechanical responses that are difficult to achieve in conventional homogeneous materials. For example, configurations like re-entrant, cross-chiral, or rotating-unit structures can induce a negative Poisson's ratio effect. This behavior is characterized by lateral expansion under tension and lateral contraction under compression [19, 25]. In addition, multistable mechanisms, progressive buckling modes, and shell-lattice or thin-walled lattice structures can enhance specific energy absorption. They can also enable programmable nonlinear mechanical responses [11, 40, 43, 49].

Extensive efforts have been devoted to characterizing and optimizing the mechanical performance of mechanical metastructures, with particular emphasis on deformation mechanisms in regular topologies, energy absorption under cyclic and impact loading, and mechanical tuning through complex structures. For regular topological structures, Wang et al. [50] combined uniaxial compression tests, numerical simulations, and large deformation theoretical analysis to reveal the yielding behavior, energy absorption characteristics, and failure modes of three-dimensional cross-chiral structures (CCS). Wu et al. [54] further examined the deformation mechanisms of three-dimensional chiral metastructures under

different loading directions and identified representative modes, including spatial rotation, tension/compression–shear coupling, and tension-induced expansion or compression-induced contraction. For cyclic and impact loading scenarios, Lu et al. [35] compared the energy absorption and shape-recovery behavior of different meta-honeycomb configurations under in-plane and out-of-plane cyclic compression, whereas Liu et al. [33] demonstrated the dual-stress-plateau response and energy-storage retention of three-dimensional reconfigurable auxetic lattices subjected to cyclic loading-unloading and drop-weight impact. Random topologies and hybrid structures have also been explored to achieve more sophisticated stiffness modulation and nonlinear mechanical responses. For example, Vorochiral random chiral metamaterials exhibit multi-step stress plateaus [48], and cylindrical quasi-zero-stiffness metamaterials show tunable nonlinear stiffness behavior [17]. In addition, Florijn et al. introduced programmable mechanical metamaterials in which lateral constraints can be adjusted to produce monotonic, non-monotonic, and hysteretic responses under uniaxial compression, demonstrating the geometric programmability of nonlinear mechanical behavior [15]. Overall, these studies have substantially advanced the understanding of deformation, failure, and energy absorption behavior in standalone mechanical metastructures with different topologies and loading conditions. However, most existing studies still treat metastructures as standalone bodies or nearly unconstrained structures. Their boundary conditions are typically simplified as free, weakly constrained, or idealized rigid boundaries, which cannot fully capture the mutual compression and lateral confinement between the metastructure skeleton and the surrounding deformable medium [13, 37, 39, 56]. In terms of loading paths, existing tests have mainly focused on uniaxial compression, in-plane or out-of-plane cyclic compression, or single impact loading. Such simplified paths are insufficient to characterize the path-dependent responses associated with changes in confining pressure, deviatoric stress adjustment, and non-proportional loading–unloading processes in engineering applications. Consequently, the strength, deformation, and energy absorption mechanisms of composite systems formed by the interaction between mechanical metastructures and granular media remain insufficiently understood, especially under lateral confinement and complex stress paths.

To address this gap, this study investigates a coupled system composed of mechanical metastructures and granular media, with re-entrant auxetic metastructures selected as the representative skeleton topology. Specifically, when re-entrant auxetic metastructures are embedded in granular assemblies to form granular composites, the contact constraints and stress transfer processes between the metastructure and the granular medium may collectively alter the strength, deformation stability, and energy absorption behavior of the composite under different stress paths. To accurately realize the target re-entrant auxetic geometry and obtain flexible metastructure skeletons with well-defined morphology and stable fabrication quality, thermoplastic polyurethane (TPU) elastomeric metastructure was fabricated using additive manufacturing [16, 33, 45]. The metastructure was then embedded in granular assemblies to form a structure–granular composite that combines the discreteness of granular media with the continuity of the structural skeleton. On this basis, the deformation response and energy absorption characteristics of the composite under different stress paths were investigated. To reveal the macroscopic mechanical response and energy absorption behavior of the composite under representative laterally confined stress paths, an axisymmetric stress-path control system was used to conduct three types of tests: monotonic loading, cyclic loading, and constant shear stress loading. Specifically, the monotonic loading tests, including constant- σ'_3 and isochoric shear paths, were used to evaluate the effects of the metastructure on the shear strength, stress-state evolution, and axial deformation of the composite. Strain-controlled cyclic loading tests were performed to characterize the hysteretic response, damping-ratio variation, residual axial strain accumulation, and evolution of energy absorption capacity of the composite during repeated loading. The constant shear stress tests were further used to examine the path-dependent deformation response and resistance to deformation instability of the granular composite when the deviatoric stress was maintained constant while the effective confining pressure was adjusted. In addition, the energy absorption performance of different metastructure arrangements was compared, and a mass-normalized specific net energy index, w_{net} , was introduced to evaluate the practical energy absorption efficiency while accounting for the structural mass penalty. Preliminary results indicate that the spatially overlapping constraints between multiple metastructure

layers may induce an energy-efficiency crossover phenomenon. This finding provides a new perspective for understanding the multiscale energy absorption mechanisms of granular composites.

2. Test Material and Setup

2.1 Test Material

The granular material employed in this study was Fujian standard sand, a well-established granular testing material in civil engineering. It is primarily composed of quartz, with a stable mineral composition and low impurity content. In addition, its stable source and relatively low cost make it particularly suitable for repeated and large-scale laboratory testing. The particle size distribution (PSD) of the granular material is presented in Figure 1, which indicates that the tested material exhibits a relatively poor gradation, based on ASTM D2487-17e1 [5].

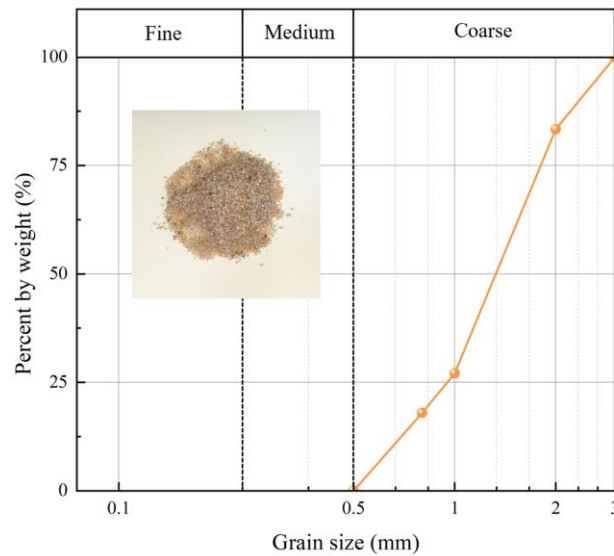


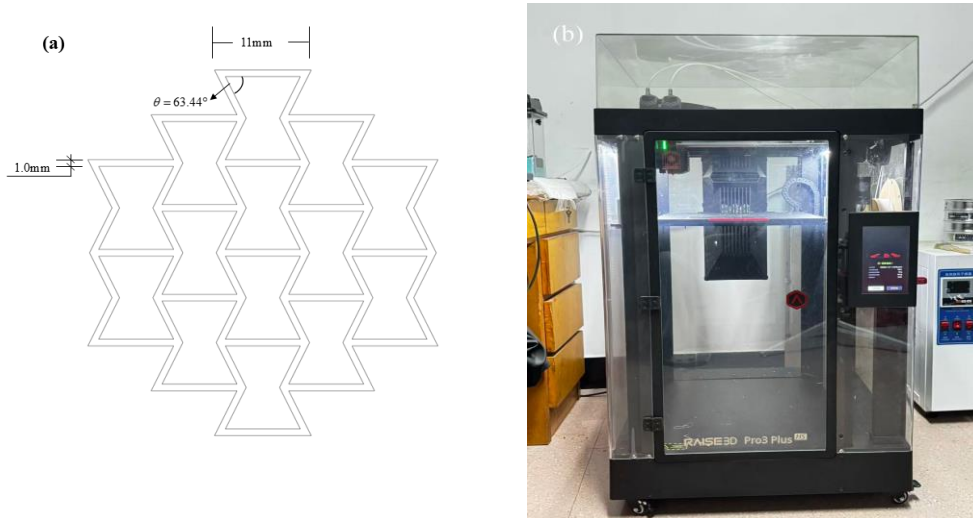
Figure 1. Particle-size distribution of Fujian standard sand.

The specific gravity (G_s) of the sand was measured as 2.69 as per ASTM C127-15 [4], with a coefficient of uniformity (C_u) of 2.376 and a coefficient of curvature (C_c) of 1.585 according to ASTM C136-06 [2]. A summary of the other basic physical properties based on the ASTM standard is provided in Table 1.

Table 1. Physical properties of Fujian sand.

Property	Value
Specific G_s	2.690
Maximum void ratio $e_{\max}$	0.732
Minimum void ratio $e_{\min}$	0.557
Medium particle size d_{50} (mm)	1.408
Coefficient of uniformity C_u	2.376
Coefficient of curvature C_c	1.585
Relative density D_r	30% / 70%

158 The metastructure used in this study was additively manufactured with a negative
 159 Poisson's ratio (NPR). The detailed geometric parameters of this specially designed auxetic
 160 honeycomb structure are explicitly presented in Figure 2(a). A detailed discussion of the
 161 structural mechanism can be found in Sun et al. [46]. All the specimens were 3D-printed
 162 from thermoplastic polyurethane (TPU) using the equipment shown in Figure 2(b).

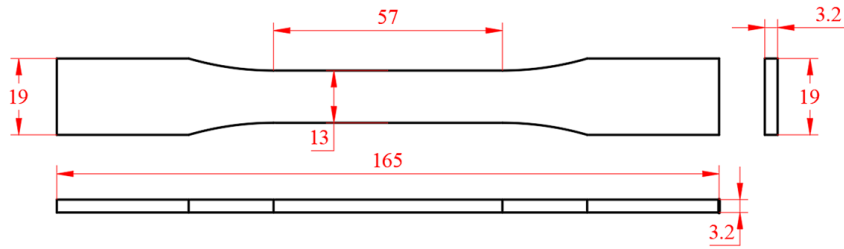


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Figure 2. (a) Geometric photograph of metastructure and (b) picture of 3D printer equipment

165 The key macroscopic structural parameters and inherent material properties of the
 166 metastructure are summarized in Table 2. To accurately determine the inherent base polymer
 167 properties listed in Table 2, independent standard uniaxial tensile tests were conducted in
 168 accordance with ASTM D638 [3]. As shown in Figure 3(a) and 3(b), Type I dumbbell-shaped
 169 strip specimens were fabricated using the exact same printing parameters as the model
 170 metastructures. The resulting engineering stress-strain curve (Figure 3c) illustrates the highly
 171 hyperelastic nature of the TPU material, providing an initial Young's modulus (E_g) of 128.6

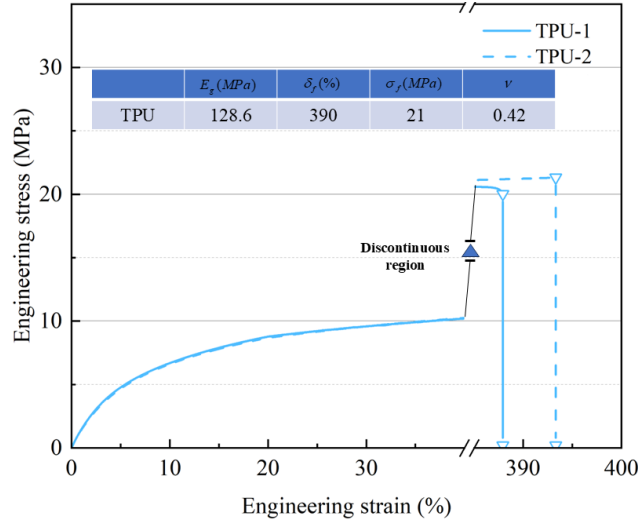
MPa and an ultimate tensile strength (σ_f) of 21 MPa. Moreover, the material exhibits an exceptional elongation at break of 390% and an inherent material Poisson's ratio of 0.42.



(a) Geometric picture of dumbbell-shaped strip



(b) Tensile test setup



(c) Tensile stress-strain behavior

Figure 3. Uniaxial tensile test of the 3D-printed TPU material: (a) Dimensions of the dumbbell-shaped strip specimen (in mm), (b) experimental setup, (c) engineering stress-strain curve.

These characterized properties confirm that the selected TPU possesses relatively low stiffness but extreme deformability, which is essential for mobilizing the structural NPR effect and absorbing cyclic energy without brittle failure.

Table 2. Physical and mechanical properties of 3D-printed metastructure

Metastructure height (mm)	Metastructure thickness (mm)	Metastructure material	Young's Modulus E_g (MPa)
10	1	TPU	128.6

2.2 Test Setup

The apparatus employed in this study is an advanced dynamic triaxial testing system developed by Geotop Technology Zhejiang Limited and is shown in Figure 4. The system mainly consists of three components. The first is the axial loading device and pressure chamber, which provides axial displacement and loading. This part is also equipped with a pore pressure transducer, as well as axial displacement and loading sensors, to monitor the variations of pore pressure, axial displacement, and load in real time during the tests, respectively. The second is the cell pressure and back-pressure system, which is controlled by

two pressure-volume controllers. The third is a data acquisition system that continuously records and processes the test data.

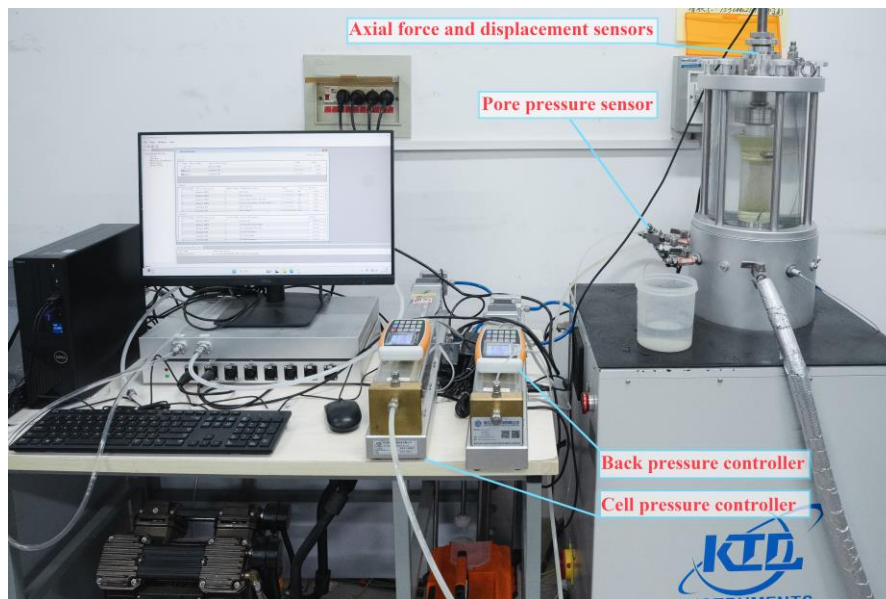


Figure 4. Test equipment

The metastructure-embedded granular composite has a diameter of 50 mm and a height of 100 mm, prepared using the pluviation method. To comprehensively investigate the mechanical behavior of the composite, granular material with two relative densities, i.e., $D_r = 30\%$ and $D_r = 70\%$, were considered. For each density, three test groups were prepared depending on whether metastructure were included and on the number of metastructure layers. The detailed grouping scheme is summarized in Table 3 in the Appendix. The placement of the metastructures is illustrated in Figure 5. In the case of a single-layer reinforcement, the metastructure was positioned approximately at the mid-height of the specimen. For the three-layer reinforcement, the specimen was divided into four sections during preparation, allowing the metastructure to be placed between every two layers of granular material. After specimen preparation, the pressure chamber was sealed in place and the saturation process was carried out. To ensure that the Skempton B-value reaches 0.95 or higher before consolidation, the specimens were first flushed with carbon dioxide for about 15 mins to displace the air. This was followed by water saturation under a hydraulic head for approximately 20 mins, and then back-pressure saturation for about 2 h, where a back pressure of 800 kPa was applied in this study. The saturation procedure was intended to

measure the volume change of the granular composite during loading, which further facilitated the assessment of energy-absorbing features of the composite.

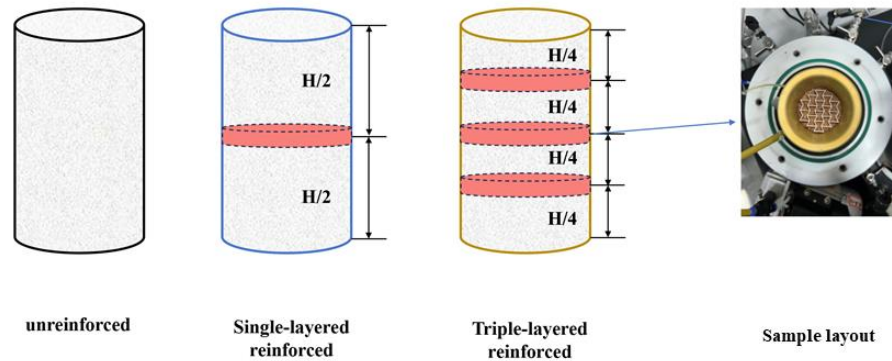


Figure 5. Schematic diagram of metastructure embedment

To comprehensively investigate the mechanical behavior of the metastructure under different confining pressures, four effective confining pressures of 100 kPa, 200 kPa, 400 kPa, and 800 kPa were applied, respectively, for consolidation and subsequent shearing. During each consolidation stage, the volumetric change of the specimen was recorded, and the corresponding void ratio after consolidation was calculated. In addition, three different loading modes were conducted. The monotonic loading under two conditions: constant-volume (isochoric) compression and constant- σ'_3 (vented) compression. For isochoric compression, the specimen was sheared with zero volumetric change, which is like the isochoric condition in soil mechanics. For vented compression, the specimen was sheared under a constant effective confining pressure with a vented boundary, allowing free changes in volume. The loading rates were 0.045 mm/min and 0.1 mm/min, respectively. These relatively low loading rates were chosen to ensure the dissipation of pore water pressure under constant- σ'_3 compression. The second was cyclic loading, which was carried out to explore the energy-absorbing behavior of the composite under repeated loading. In this test, the cyclic loading frequency was 1 Hz, with a strain amplitude of 1 mm, and 50 data points were recorded per second. This specific frequency and amplitude configuration is widely adopted as a standard reference in laboratory cyclic triaxial testing to reasonably simulate typical seismic loading conditions, while ensuring reliable excess pore water pressure equalization and measurement within the specimen [14, 52]. The third was CSS loading,

which was adopted to examine the composite performance under complex stress path. In this case, the target deviatoric stress q was maintained at 250 kPa, while two effective confining pressures of 150 kPa and 200 kPa were applied, respectively.

The detailed void ratios of the specimens after consolidation and the classification of the test groups are summarized in Table 3.

3. Results

3.1 Stress-Strain Responses under Monotonic Load

3.1.1 Mechanical Behavior under constant- σ'_3 compression

Figure 6 presents the test results of three specimens under constant- σ'_3 condition. Specifically, Figure 6(a, c, e) and Figure 6(b, d, f) show the stress–strain curves and volumetric strain–axial strain curves, respectively, for the three types of test series. In Figure 6, L-EM/D-EM denote the unreinforced groups; L-M1/D-M1 represent the groups reinforced with a single metastructure layer; L-M3/D-M3 correspond to the groups reinforced with three metastructure layers. Here, “L” and “D” indicate loose and dense initial states, respectively.

As shown in Figure 6(a, c, e), under constant- σ'_3 conditions, q increases with axial strain, reaches a peak strength, and then decreases gradually before stabilizing near the residual strength. With an increasing number of metastructure inclusions, the axial strain required to reach the peak strength becomes significantly larger. for the unreinforced specimens, the peak generally occurs at about 5% of axial strain, while for granular composites reinforced with one metastructure layer it occurs at approximately 8%, and for those with three layers it typically appears at around 10%. This indicates that metastructures can effectively delay the occurrence of peak strength, and the delaying effect becomes more pronounced with additional layers.

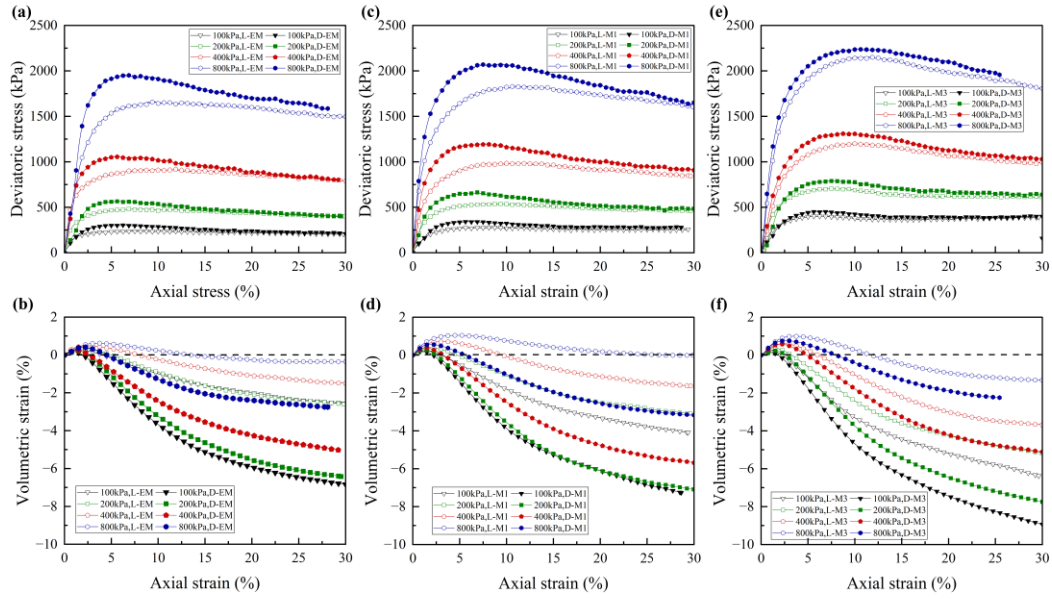


Figure 6. Constant- σ'_3 shear behavior of granular composites at different initial states.

In addition, under the same number of metastructures, the peak strength is influenced by both the initial relative density and the initial confining pressure. At a given confining pressure, higher relative density leads to greater peak strength; at a given density, the peak strength increases markedly with higher confining pressure. In contrast, the residual strength is mainly governed by the confining pressure. Specifically, at the same confining pressure, specimens with different densities exhibit nearly identical residual strengths, which is consistent with the findings of a previous study [47].

Furthermore, as shown in Figure 6(a, c, e), under the same initial density and confining pressure, both the peak and residual strengths increase with the number of metastructures. This demonstrates that the confining effect of metastructures can effectively enhance the overall strength of the granular composites, which agrees well with the conclusions reported in another study [27].

In addition to the stress–strain response, the volumetric behavior of the specimens is illustrated in Figure 6(b, d, f). Under the same number of metastructures, both dense and loose granular composites exhibit a similar pattern. the volumetric strain initially shows contraction with increasing axial strain, reaches a maximum contraction at relatively small axial strain, and then gradually transforms into dilation. Furthermore, at the same relative density, the ultimate extent of dilation decreases as the initial confining pressure increases.

This indicates that higher confining pressure imposes a stronger constraint on the volumetric deformation of the granular composites [36].

3.1.2 Behavior under Constant-volume Shear

The constant-volume shear behaviors of augmented granular composite specimens, alongside unmodified control specimens are illustrated in Figure 7. Figure 7(a, c, e) present the stress–strain relationships of specimens without metastructures, with a single metastructure, and with three metastructures, respectively. A comparison between dense and loose specimens under the same reinforcement condition reveals distinct differences in strength development. In loose specimens, strength increases continuously with axial strain and gradually approaches a stable value without exhibiting stress-softening behavior [22]; in contrast, dense specimens attain a peak strength followed by a reduction, indicating typical stress-softening behavior. Figure 7(a) further demonstrates that initial density has a marked influence on overall strength, with dense specimens consistently stronger than loose ones. By contrast, for specimens with the same density, the effect of initial confining pressure is relatively small, which contrasts with the findings under constant- σ'_3 conditions [6]. Moreover, comparison of Figure 7(a, c, e) shows that increasing the number of metastructures leads to a clear enhancement of granular composites strength under the same initial conditions, in agreement with the constant- σ'_3 test results.

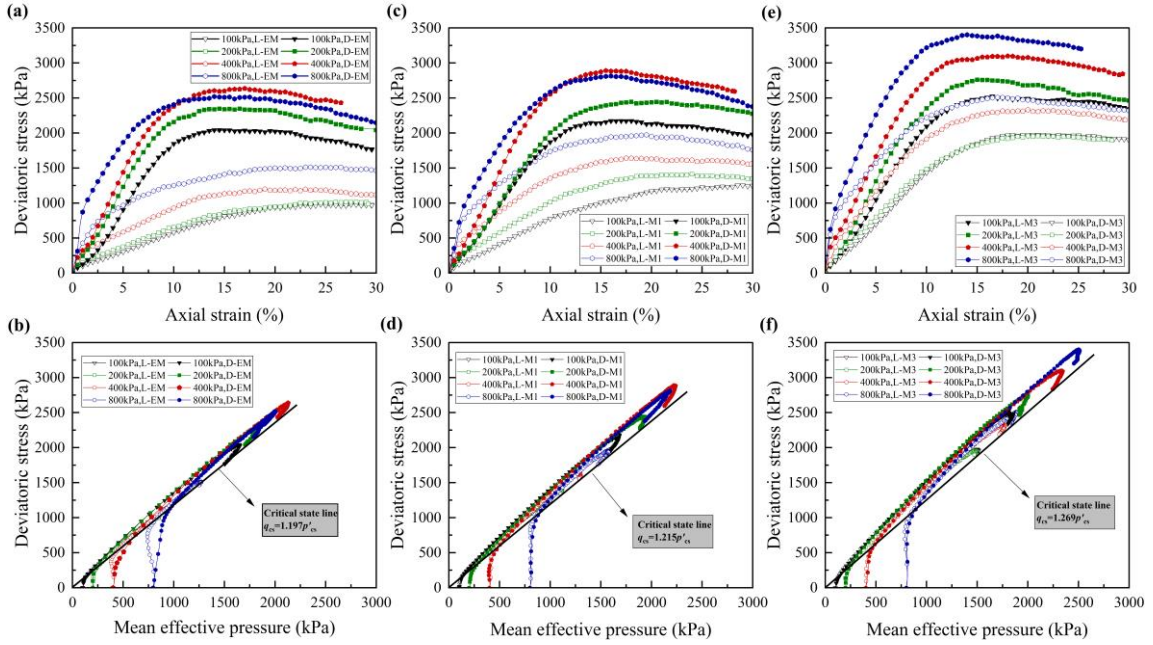


Figure 7. Constant-volume shear behavior of granular composites at different initial states.

The corresponding stress paths under constant-volume shear are shown in Figure 7(b, d, f). In Figure 7(b), all stress paths cluster around a straight line through the origin when it reaches a certain state. During this state the particle mass tends to be in a state of continuous deformation under constant shear stress (CSS) and constant mean effective stress, largely independent of initial density and confining pressure. This line is commonly referred to as the residual state line. Comparison of Figure 7(b, d, f) indicates that each specimen type corresponds to a distinct residual state line, and the slope of this line increases with the number of metastructures. This implies that at the residual state, specimens reinforced with metastructures can sustain higher strength than unreinforced ones at the same mean effective stress.

3.1.3 Mechanical Behavior under CSS

In practical terms, instability represents the loss of a composite's ability to sustain its existing stress state, resulting in the development of large plastic deformations [38]. In laboratory tests, instability is typically identified using several criteria: (i) a sudden increase in accumulated plastic deformation; (ii) a rapid reduction in deviatoric stress; or (iii) a negative second-order work [38].

To investigate the instability resistance of granular composites, a series of CSS tests were conducted on the three specimen types. The loading procedure consisted of initially following

a conventional constant- σ'_3 path until q reached 250 kPa, after which q was held constant while the mean effective stress was gradually reduced. The test results are presented in Figure 8, where two initial mean effective stress levels were adopted. Instability was identified at the point of rapid deviatoric stress reduction, marked with star symbols in Figure 8.

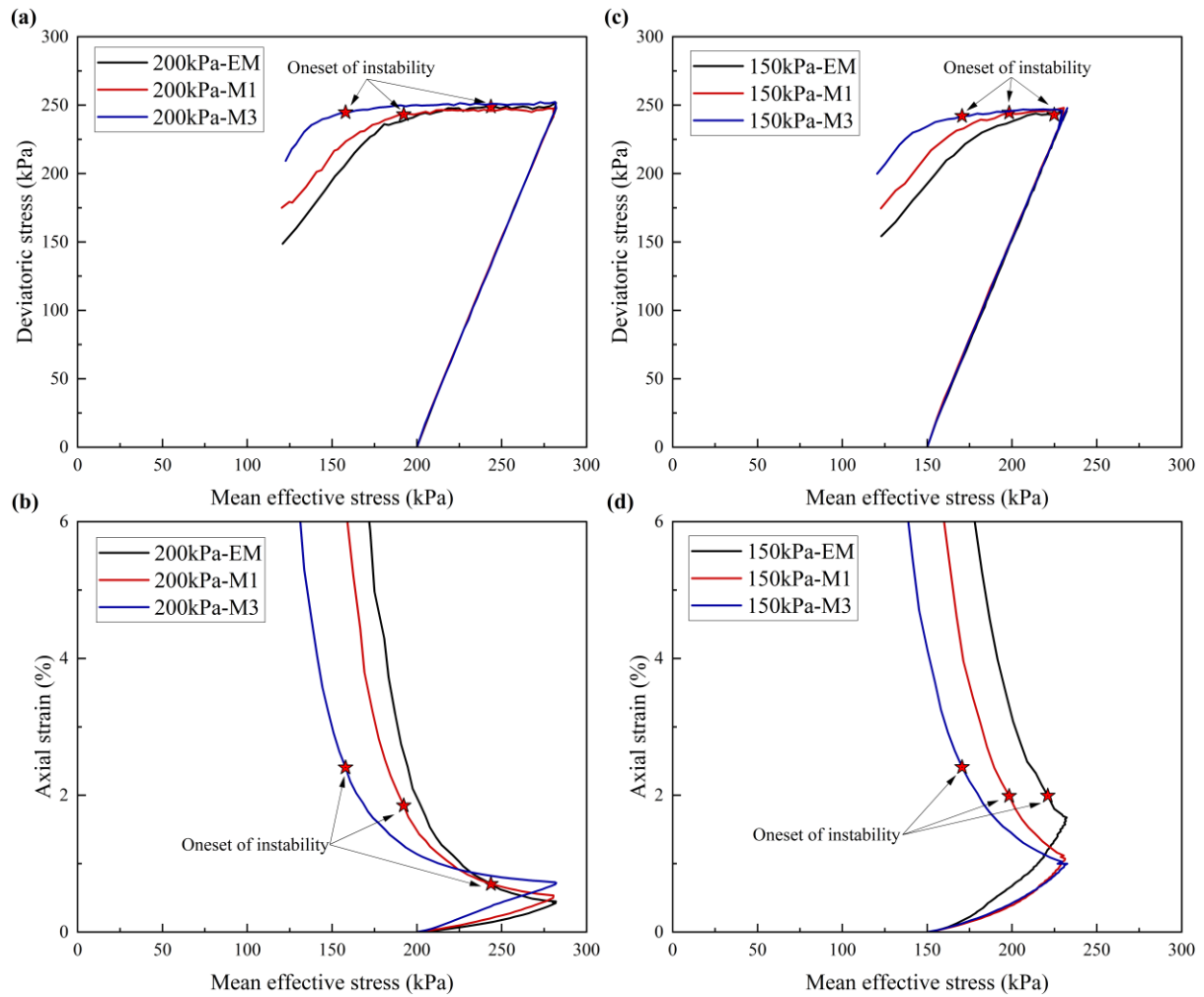


Figure 8. CSS behavior of granular composites at different initial effective mean pressure: (a)(c) stress paths, (b)(d) axial strain versus mean effective pressure.

A comparison of the results shows that, in the stress paths of Figure 8(a, c), the curves of metastructure-reinforced specimens consistently lie above those of unreinforced ones, indicating that reinforcement substantially improves specimen stability and delays the onset of instability. In the axial strain–mean effective stress plots of Figure 8(b, d), the reinforced specimen curves lie below those of unreinforced ones, demonstrating that metastructure imposes significant constraint on axial deformation and postpones the stage at which

instability occurs. These findings confirm that metastructure effectively enhances the instability resistance of granular composites, and that the reinforcing effect becomes more pronounced with an increasing number of metastructures.

3.2 Characteristic State Behaviors

3.2.1 Residual-State

To ensure that the specimens would reach the residual state during monotonic loading, the axial strain at the end of all tests was fixed at 30%. However, not all specimens attained the residual state. For instance, in the unreinforced group, some dense specimens exhibited pronounced dilation during loading, while the relatively low confining pressure was insufficient to suppress this volumetric expansion, ultimately preventing the specimens from reaching a stable residual state, as also reported by [18]. Therefore, only data from specimens that reached the residual state were used to determine the residual state line.

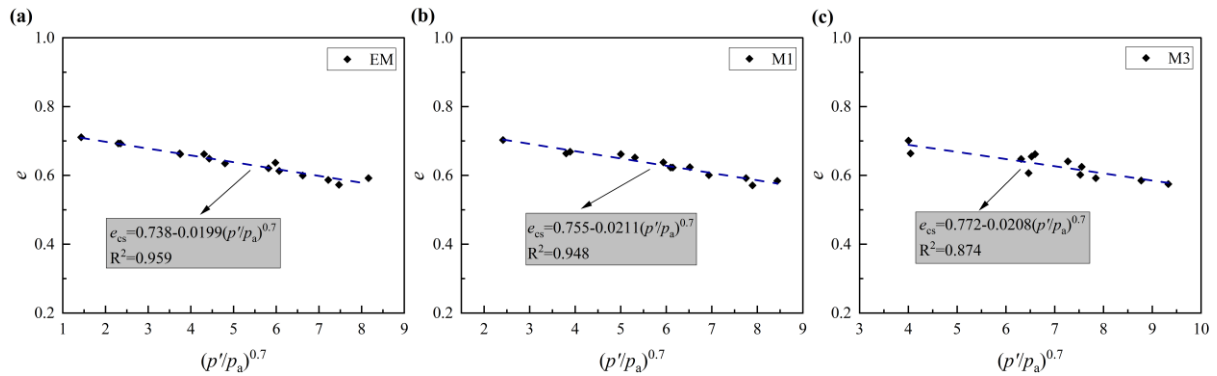


Figure 9. Residual-state lines of different types of specimens in $e - (p' / p_a)^{0.7}$ plane.

Figure 9 shows the residual state locations of the three specimen types in the $e - (p' / p_a)^{0.7}$ plane, normalized with respect to $p_a = 101$ kPa. The results indicate that the residual state points for all specimen types lie approximately on straight lines and are well correlated. With increasing metastructure, the corresponding residual state lines shift towards higher void ratios and mean effective stresses. This suggests that, at an identical residual void ratio, the reinforced granular composites sustain markedly higher mean effective stresses than the unreinforced ones, confirming the beneficial effect of metastructure reinforcement on composite strength at the residual state.

Figure 10 shows the effect of metastructures on the residual stress ratio in the $q - p'$ plane. It is observed that, with increasing metastructures, the slope of the residual state line and hence the residual stress ratio increases [51]. This finding suggests that, for a given mean effective stress, the reinforced granular composites can sustain higher residual deviatoric stresses, thereby demonstrating the role of metastructures in enhancing the shear resistance of granular composite at residual state.

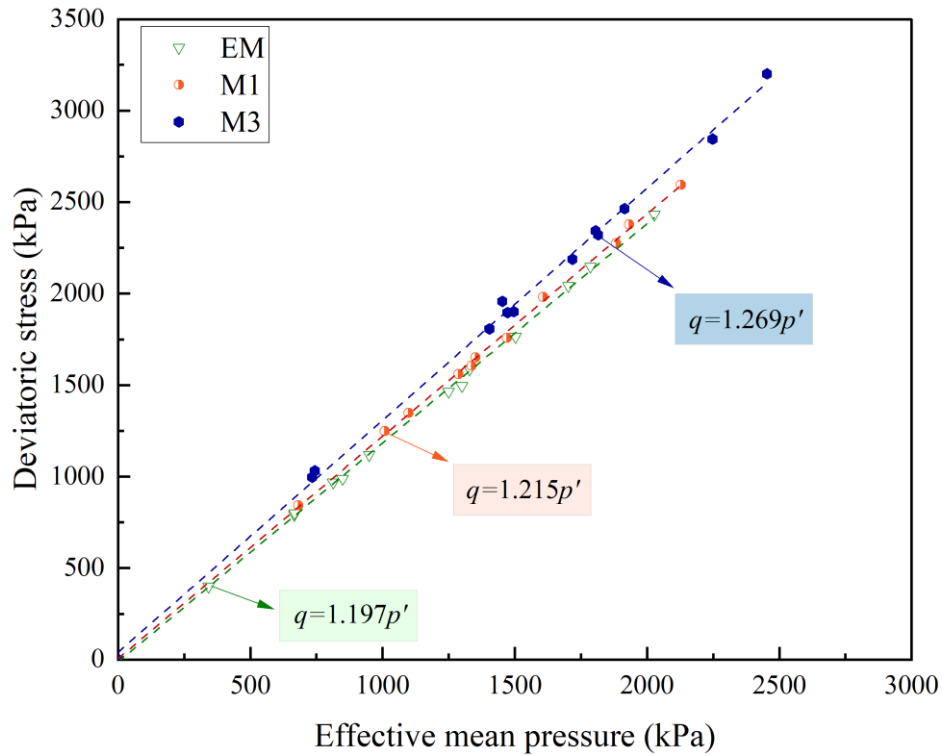


Figure 10. Residual-state lines of granular composite in $p' - q$ plane.

3.2.2 Phase-Transformation State

The volumetric response of granular composites during loading, whether under constant- σ'_3 or constant-volume conditions, is of considerable significance. Such volumetric behavior is typically characterized by two stages: contraction and dilation. The phase-transformation state indicates that the rate of volumetric contraction transits to the rate of volumetric dilation [26]. At this state $d\varepsilon_v^p / d\varepsilon_a^p = 0$, where $d\varepsilon_v^p$ and $d\varepsilon_a^p$ denote the increments of plastic volumetric strain and plastic shear strain, respectively. In this study, the phase-transformation point was identified as the peak of the volumetric strain–axial strain curve in free-volume

shear tests, and as the peak of the pore water pressure–axial strain curve in constant-volume shear tests. These identified points were then used to extract and calculate the relevant data for subsequent analysis.

Figure 11 illustrates the relationship between the stress ratio at the phase-transformation state and the state parameter for different granular material. The state parameter is employed to quantify the proximity of granular assembly to the residual state, and is defined as the difference between the initial void ratio and the residual-state void ratio under the same mean effective stress [8]. The detailed calculation of the stress ratio at the phase-transformation state (M_d) and the state parameter (ψ) is provided in the Appendix. As shown in the Figure 11, M_d at the phase-transformation state exhibits a positive correlation with the state parameter across all tested specimens, with data points clustering closely around a single straight line. A fitting function was therefore applied to describe this relationship [29].

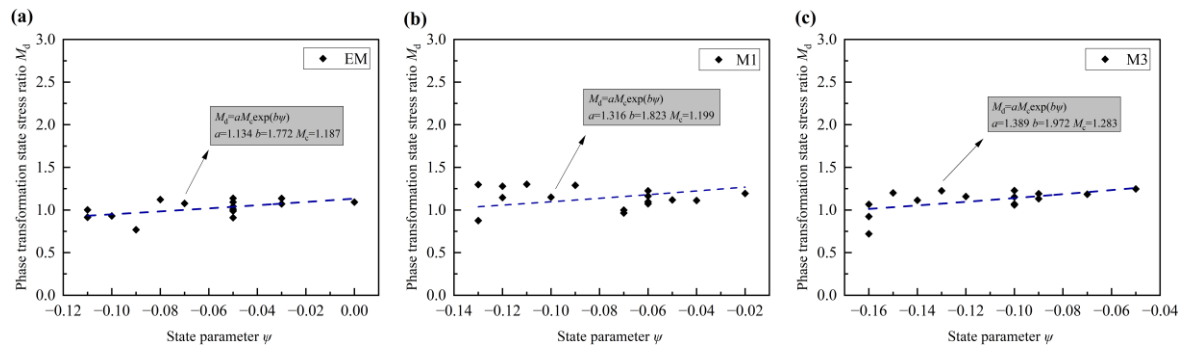


Figure 11. Phase-transformation state stress ratio versus state parameter

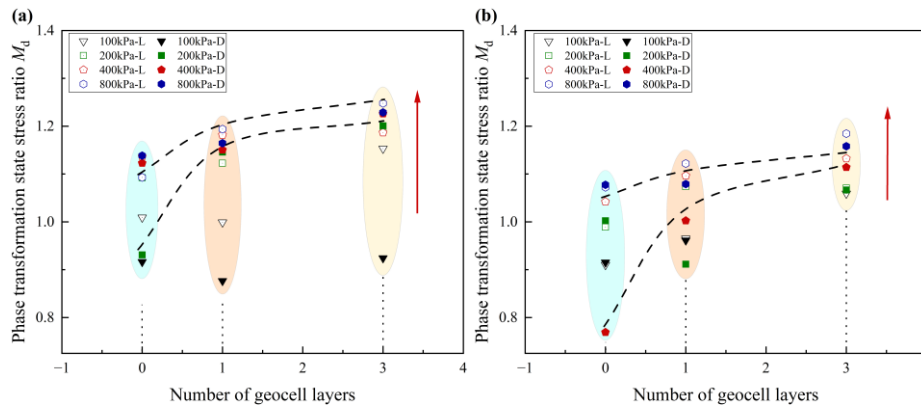


Figure 12. Phase-transformation state stress ratio versus reinforcement number under (a) constant- σ'_3 conditions and (b) constant-volume conditions

Figure 12 presents the effect of metastructures on M_d under both constant- σ'_3 and constant-volume shearing, where a reinforcement number of zero denotes unreinforced specimens. The results reveal two key observations. First, M_d generally increases with confining pressure, irrespective of specimen type. Second, M_d tends to rise with an increasing number of reinforcement layers, demonstrating the beneficial role of metastructures in enhancing M_d . A further comparison between monotonic constant- σ'_3 and constant-volume shear conditions indicates broadly consistent trends, although the reinforcing effect appears to be slightly more pronounced under constant-volume shear conditions.

3.2.3 Peak-State

In addition to the residual and phase-transformation states, the peak state during loading is also of considerable significance. Accordingly, the experimental data at the peak state were analyzed and summarized in this study to further investigate the role of metastructure.

Figure 13 presents the relationship between the stress ratio and the state parameter at the peak state for different specimens. The results indicate that, regardless of specimen type, the peak-state stress ratio M_p exhibits a negative correlation with the state parameter, with data points distributed in a well-defined manner and fitting closely to the regression curve. A further comparison shows that increasing the number of metastructures causes the fitted curve to shift upward and to the right, implying that at the same state parameter, specimens with more reinforcement layers attain higher M_p .

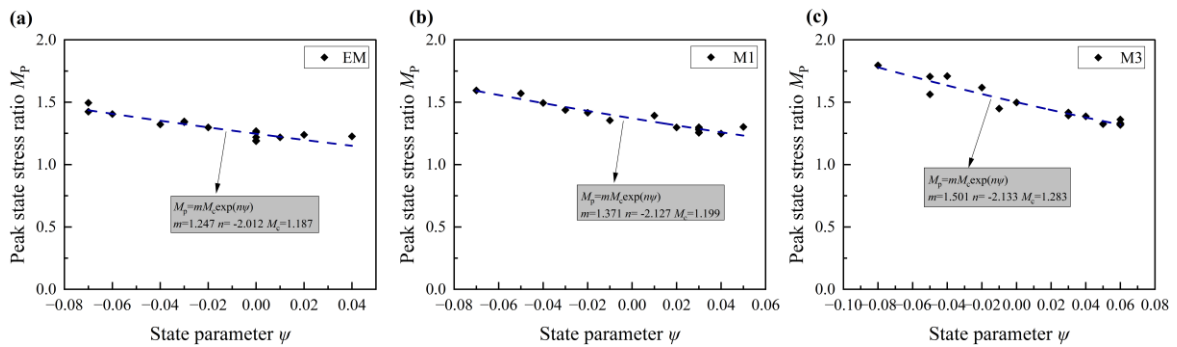


Figure 13. Peak-state stress ratio versus state parameter

Figure 14 illustrates the variation of M_p with metastructures under both monotonic constant- σ'_3 and constant-volume shear conditions. The results demonstrate that metastructure inclusion leads to a pronounced increase in M_p under both monotonic constant- σ'_3 and constant-volume shear conditions, underscoring the effectiveness of metastructures in enhancing peak strength. However, unlike the phase-transformation state, M_p generally decreases with increasing confining pressure, and this trend appears to be independent of reinforcement and initial density.

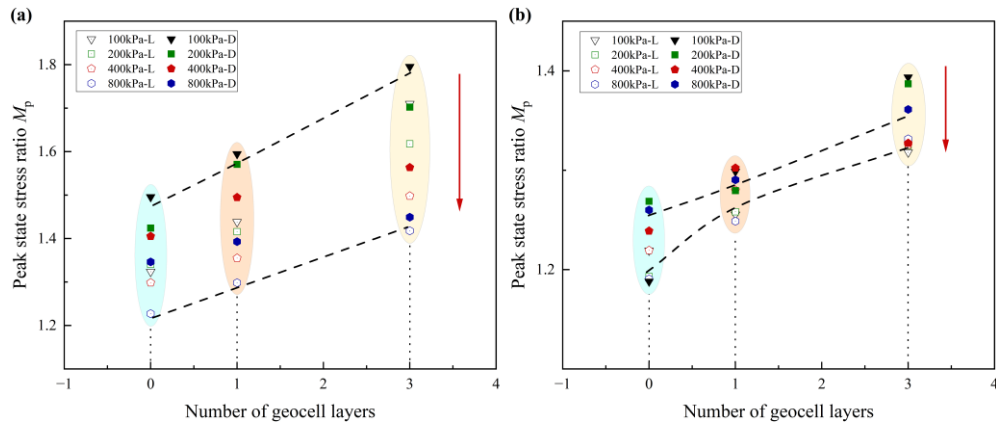


Figure 14. Peak-state stress ratio versus metastructure under (a) constant- σ'_3 and (b) constant-volume shear condition

3.3 Behavior with Strain Cycles under Constant-volume Shear

To comprehensively evaluate the deformation and energy-absorbing features of the composites, a series of constant- σ'_3 and constant-volume cyclic triaxial tests were performed. EM, M1, and M3 correspond to the unreinforced, single-layer metastructure embedded, and three-layer metastructures embedded groups, respectively. To ensure that all specimens reached instability during loading, the initial relative density was controlled at 30%, with cyclic displacement amplitude and frequency set to 1 mm and 1 Hz, respectively.

3.3.1 Stress Strain Response

Figure 15 presents twelve sets of deviatoric stress-axial strain curves obtained for the three types of specimens under different initial confining pressures. Figure 15(a-d) correspond to the unreinforced group, where the hysteresis loops gradually contract during cyclic loading, and the deviatoric stress eventually falls to zero, indicating instability. With increasing confining pressure, the peak deviatoric stress rises accordingly, consistent with previous findings [52]. Through comparison, one can further observe that at an identical initial confining pressure, the peak deviatoric stress increases with the number of metastructures, demonstrating that metastructure inclusion markedly enhances composite strength. Moreover, metastructure-embedded granular composite exhibit tighter hysteresis loops, a feature that becomes more pronounced with additional metastructure layers, reflecting stronger resistance to cyclic disturbance and lower instability susceptibility [9].

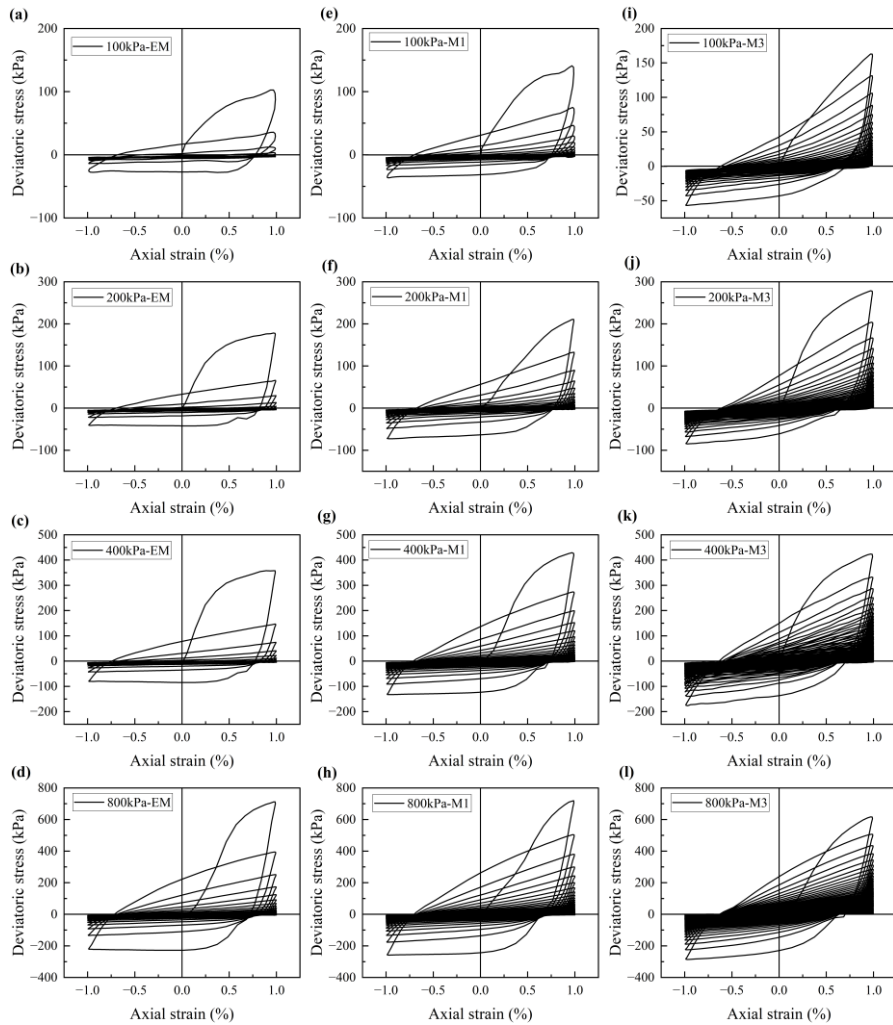


Figure 15. Stress-strain relationship measured in constant-volume cyclic triaxial tests

Figure 16 shows the effective stress paths of the three specimen types under different initial confining pressures. During cyclic loading, excess pore water pressure gradually accumulates, reducing effective stress until it dissipates completely and instability occurs. For the unreinforced specimens Figure 16(a-d), as the initial confining pressure increases from 100kPa to 800 kPa respectively, the stress paths change slightly. First, the initial dilative tendency is suppressed; second, the stress paths become more concentrated. Comparison with the reinforced specimens under the same conditions further confirms the beneficial effect of metastructure reinforcement in enhancing instability resistance.

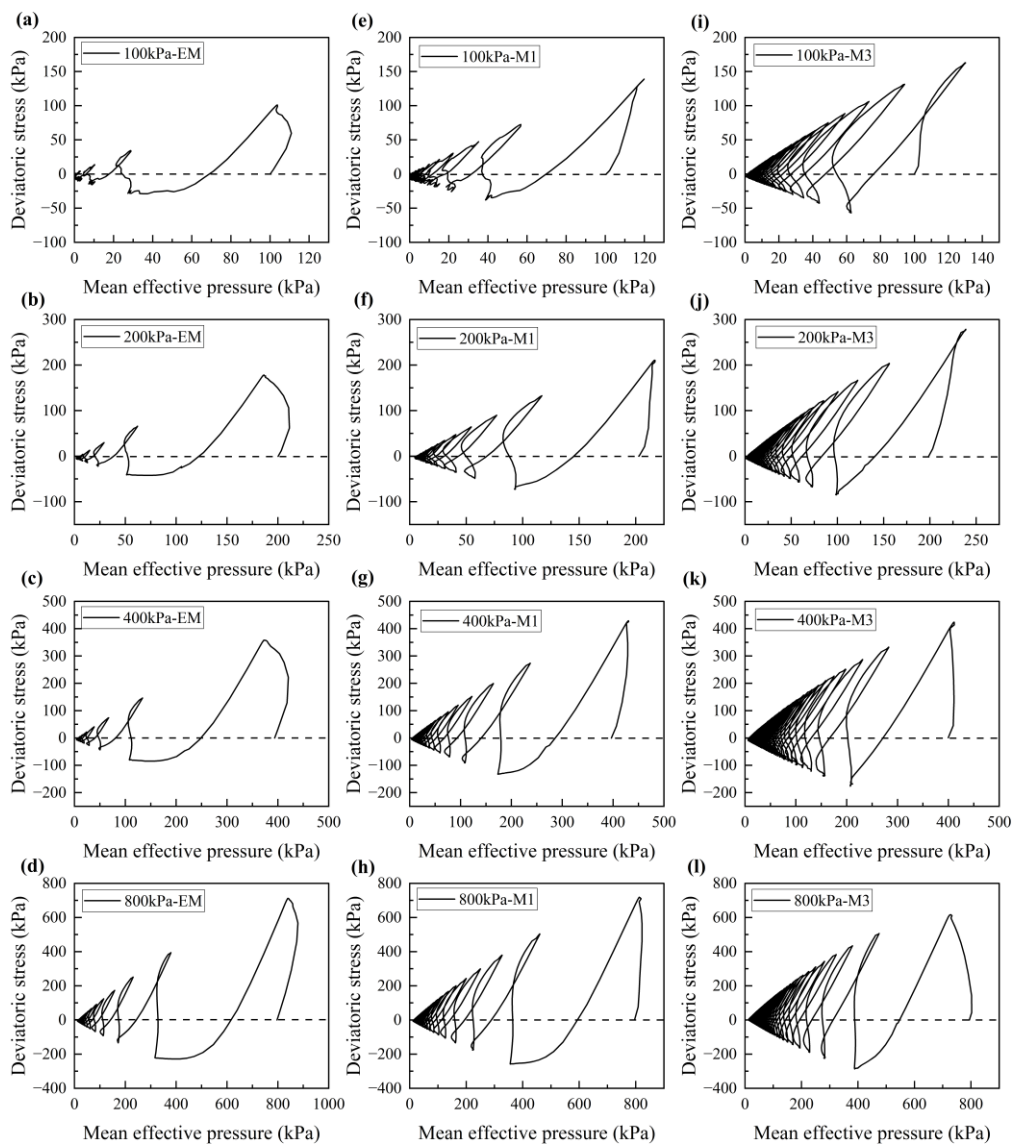


Figure 16. Effective stress paths measured in constant-volume cyclic triaxial tests

3.3.2 Accumulation of Pore Water Pressure

Figure 17 illustrates the evolution of the ratio of accumulated excess pore water pressure (Δu) to the initial mean effective stress as a function of the number of cycles under different initial confining pressures until instability, at which point loading was terminated. Various criteria have been proposed for defining instability; in this study, a practical threshold of $\Delta u / p_0' = 0.95$ was adopted to define instability, accounting for system compliance and the inherent cyclic mobility of granular composite that typically prevent the measured $\Delta u / p_0'$ from reaching absolute unity [23] and the instability is identified when the excess pore water pressure rose rapidly and the mean effective stress dissipated to zero ($q = p' = 0$). As shown, with increasing cycles, excess pore water pressure accumulates progressively until instability failure. Notably, at identical initial confining pressure, the reinforced granular composites exhibit consistently lower curves than the unreinforced ones, indicating that metastructure can effectively delay pore water pressure build-up and hence improve instability resistance. In addition, increasing the initial confining pressure significantly enhances the instability resistance of all three specimen types.

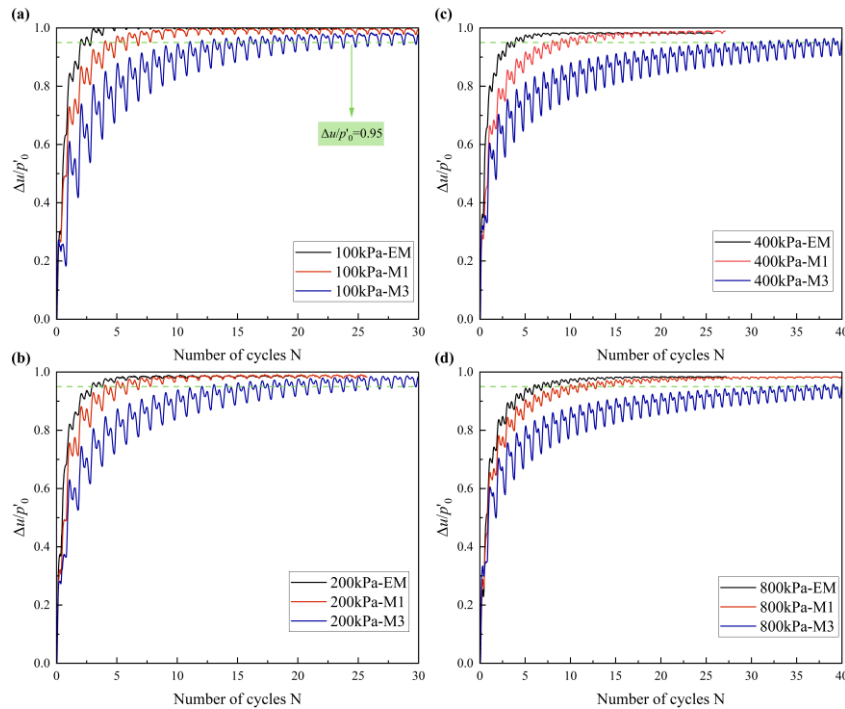


Figure 17. Ratio of accumulated pore water pressure Δu and initial mean effective pressure as a function of the number of cycles measured in constant-volume cyclic triaxial tests

3.4 Energy Absorption and Damping Ratio

3.4.1 Energy Absorption Capacity

This section investigates the energy absorption behavior of metastructure-embedded granular composite. In triaxial testing, the overall mechanical response can be effectively characterized by the energy absorbed during loading [14, 24, 55]. This energy absorption capacity can be quantified as W , defined as the cumulative dissipated energy per unit volume by the specimen during loading.

The distribution of energy differs between constant- σ'_3 and constant-volume conditions, thus the calculation of W also varies. In constant- σ'_3 tests, W_1 represents the energy associated with volumetric strain due to mean effective stress; in constant-volume tests, as no volumetric change occurs ($W_1 = 0$), only W_2 , the energy corresponding to shear strain induced by q , is considered. Specifically, W_1 is determined from the area enclosed by the mean effective stress–volumetric strain curve ($p' - \varepsilon_v$), considering only the compressive branch, whereas W_2 is obtained from the area bounded by the deviatoric stress–axial strain curve ($q - \varepsilon_a$). The detailed formulations are given in Eqs. (1–3) [12, 42], where k denotes the number of increments recorded during loading. To comprehensively evaluate the role of metastructure in modifying the composite's energy absorption capacity, results from monotonic constant- σ'_3 and constant-volume tests, and cyclic constant-volume tests are compared and discussed in detail.

$$W = W_1 + W_2, \quad (1)$$

$$W_1 = \sum_{i=1}^{k-1} \frac{1}{2} [(p')^i + (p')^{i+1}] (\varepsilon_v^{i+1} - \varepsilon_v^i), \quad (2)$$

$$W_2 = \sum_{i=1}^{k-1} \frac{1}{2} (q^i + q^{i+1}) (\varepsilon_a^{i+1} - \varepsilon_a^i). \quad (3)$$

Figure 18 illustrates the variation of the total dissipated energy per unit volume with axial strain for the three specimen types under monotonic constant- σ'_3 loading ($p_a = 101$ kPa).

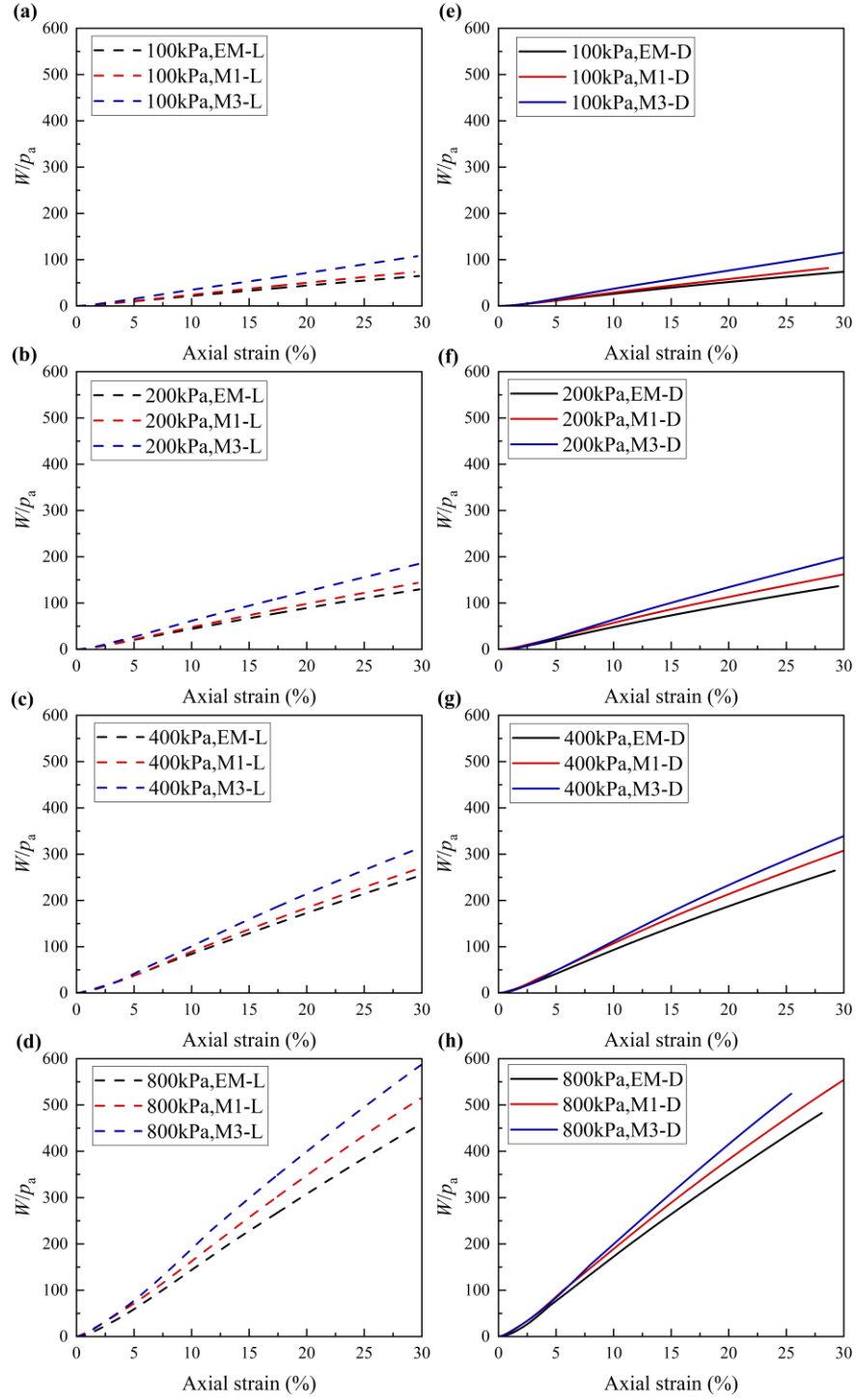


Figure 18. Cumulative dissipated energy per unit volume W / p_a versus axial strain at different initial mean effective pressure under constant- σ'_3 condition.

Here, EM, M1, and M3 denote unreinforced specimens, composites reinforced with one metastructure layer, and those reinforced with three layers, respectively; L/D differentiates loose and dense states. The results show that, regardless of initial state, metastructure

reinforcement markedly enhances energy absorption capacity, and this effect becomes more pronounced as the number of layers increases. Under identical conditions, dissipated energy increases with higher initial confining pressure. However, under constant- σ'_3 loading, the influence of density is not substantial, which is consistent with the stress-strain behavior in Figure 6. The areas enclosed by the curves for specimens of different densities but identical reinforcement is relatively similar.

Figure 19 presents the monotonic constant-volume test results. Similar to the constant- σ'_3 case, the total dissipated energy per unit volume increases with the number of metastructures. The key distinction lies in the governing role of density. Under constant-volume conditions, the final dissipated energy exhibits only limited dependence on the initial confining pressure but is highly sensitive to density. For specimens with identical confining pressure and reinforcement, higher density consistently yields greater dissipated energy. This contrast arises from the differing mechanisms: in constant- σ'_3 loading, where no excess pore-water pressure builds up, the ultimate strength is primarily controlled by initial confining pressure; in constant-volume loading, excess pore-water pressure develops due to dilatancy, and its magnitude is governed by the initial density. Consequently, the ultimate strength is mainly determined by density [31].

Figure 20 depicts the evolution of dissipated energy with loading cycles for the three specimen types, all prepared at 30% relative density. Four initial confining pressures are compared, and in all cases, the number of liquefaction cycles (N_{liq}) does not exceed 30. The results indicate: (i) consistent with monotonic tests, metastructure reinforcement substantially enhances energy absorption irrespective of initial conditions, with improvements more pronounced for greater layer numbers; (ii) the total dissipated energy increases with confining pressure, rising markedly as confining pressure increases from 100 to 800 kPa respectively; (iii) the relative benefit of reinforcement becomes increasingly significant at higher confining pressures. To explain this behavior, we hypothesize a confinement-induced ‘pre-tightening’ effect. Under elevated confining pressures, the auxetic metastructure actively contracts

inward, severely enhancing the mechanical interlocking and interfacial friction at the re-entrant nodes, thereby dissipating significantly more energy during cyclic shearing.

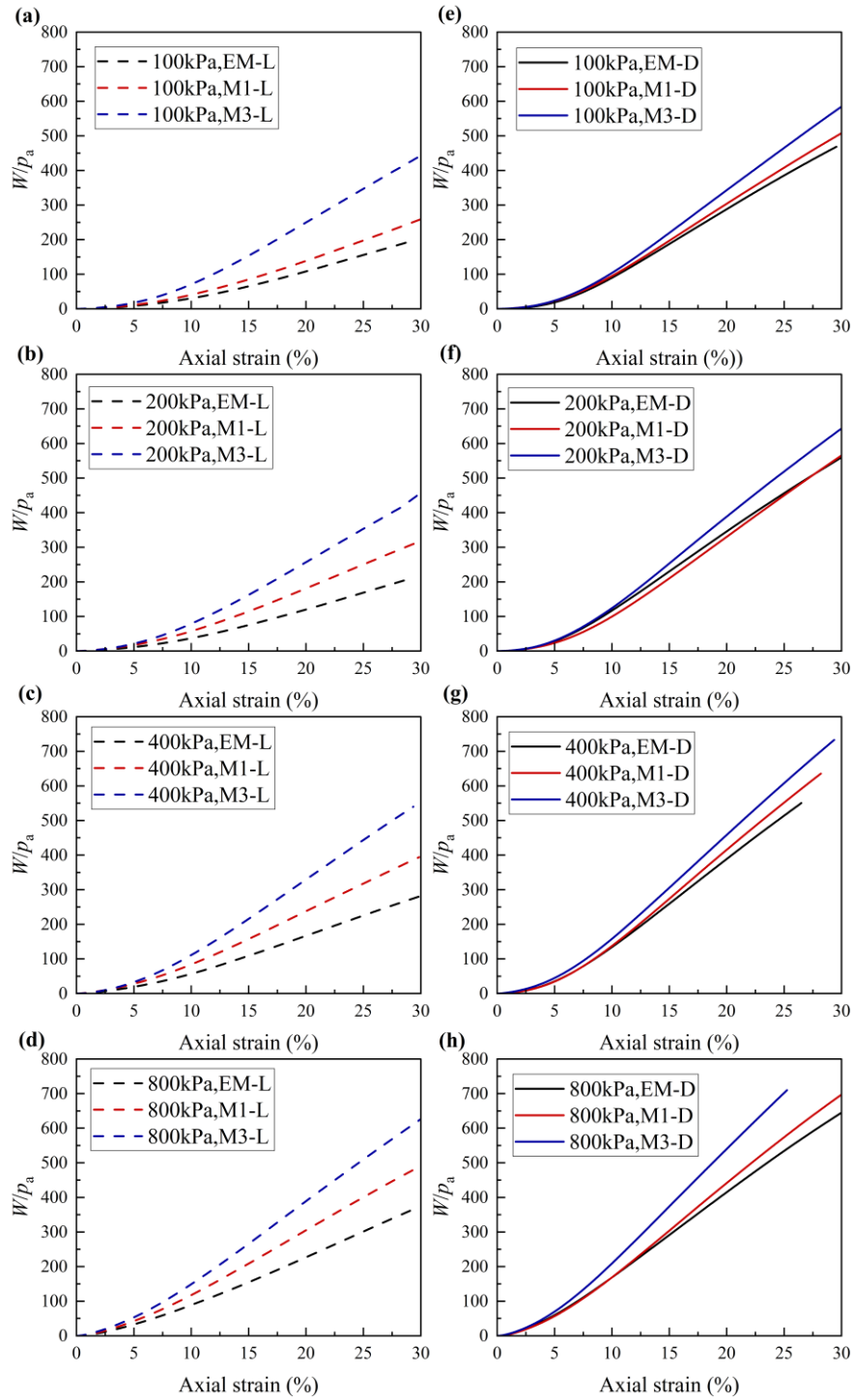


Figure 19. Cumulative dissipated energy per unit volume W / p_a versus axial strain at different initial mean effective pressure under constant-volume condition.

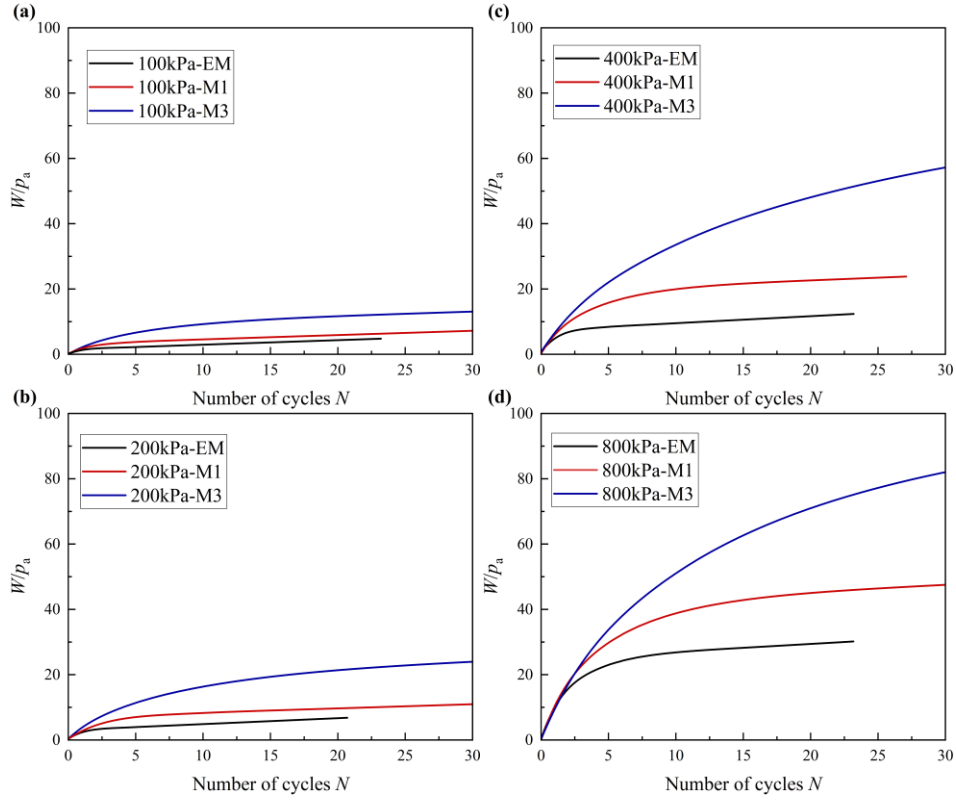


Figure 20. Cumulative dissipated energy per unit volume W / p_a versus number of cycles N at different initial mean effective pressure under constant-volume condition

While the evolution of the total cumulative dissipated energy per unit volume (Figure 20) clearly shows the overall energy absorption capacity of the specimens, this value does not consider the mass of the reinforcement used. To objectively evaluate the energy absorption efficiency per unit mass of the metastructure and allow for a fair comparison among different configurations, a mass-normalized specific net energy index, denoted as w_{net} (unit: g^{-1}), is introduced. To maintain consistency with the dimensionless analysis above, this index isolates the net energy density ratio provided by the metastructure and divides it by the actual mass of the embedded metastructure (m_m):

$$w_{\text{net}} = \frac{(W / p_a)_{\text{reinforced}} - (W / p_a)_{\text{unreinforced}}}{m_m} \quad (4)$$

where the numerator represents the net energy absorption purely contributed by the metastructure. Specifically, it is calculated as the difference between the reinforced and

unreinforced specimens tested under the exact same initial conditions (i.e., identical initial confining pressure and relative density).

Figure 21 presents the evolution of this specific net energy index for M1 and M3 configurations under initial confining pressures of 100–800 kPa respectively.

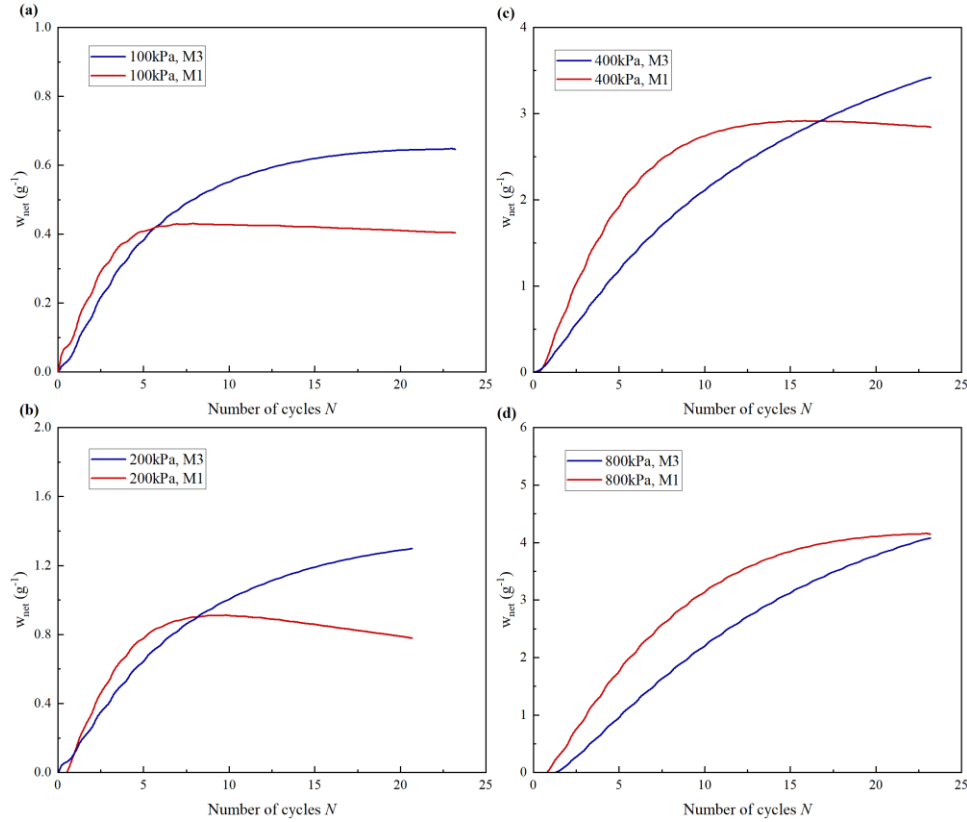


Figure 21. Evolution of the mass-normalized specific net energy index w_{net} for single-layer (M1) and triple-layer (M3) metastructure-reinforced specimens under initial confining pressures of: (a) 100 kPa; (b) 200 kPa; (c) 400 kPa; and (d) 800 kPa

An interesting efficiency crossover phenomenon is observed. During the early loading cycles, the single-layer metastructure (M1) generally shows a higher per-unit-mass efficiency than the triple-layer (M3). Since the metastructure layers in M3 are spaced apart, this initial reduction in efficiency is primarily attributed to the overlapping of their confinement influence zones. This overlapping effect stiffens the granular assembly matrix and restricts global radial deformation, preventing individual metastructure layers in M3 from stretching as fully as a single, isolated layer. However, as the cyclic loading continues, the M1 curve gradually levels off (and even slightly decreases, as seen at 200 kPa). In contrast, the M3 curve maintains a continuous increase and eventually overtakes M1 at later stages (clearly

observed at 100, 200, and 400 kPa). This delayed efficiency crossover is closely related to the occurrence of granular assembly liquefaction. The M1 composites show limited volumetric confinement, which leads to a faster accumulation of pore water pressure and earlier initial instability. Once reaching instability, the composite loses its strength, causing the metastructure to lose its effective interfacial friction, and thus the energy absorption stops increasing. Conversely, the strong 3D network of the M3 configuration significantly restricts granular assembly deformation and delays the generation of pore pressure. Therefore, M3 maintains effective particle-metastructure interaction and continues to dissipate energy long after M1 has failed. This demonstrates the important role of multi-metastructure reinforcement in maintaining long-term dynamic stability.

3.4.2 Damping ratio

The damping ratio is a fundamental indicator of dynamic response of composites, quantifying the efficiency of energy absorption under cyclic loading. It is defined in Eq. (4), where A_L is the energy dissipated within a single cycle, represented by the hysteresis loop area, and A_s is the corresponding ideal elastic strain energy, equal to the triangular area in Figure 22. More details about the calculation can be found in Sitharam and Vinod [44].

$$D = \frac{A_L}{4\pi A_s} \quad (5)$$

Figure 23 presents the variation of damping ratio with cycle number for specimens under different initial confining pressures.

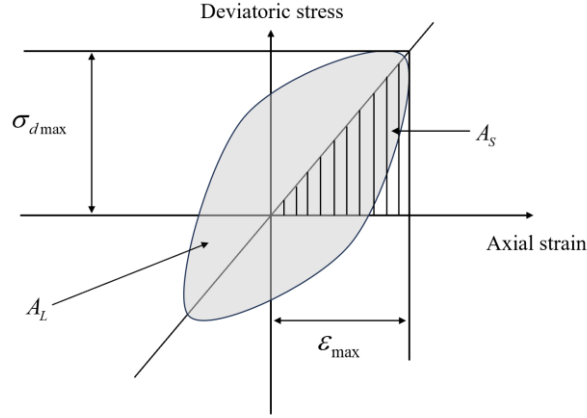


Figure 22. The area of A_L and A_S .

Results show that, at identical confining pressures, unreinforced specimens exhibit consistently higher damping ratios than reinforced ones, although the difference diminishes progressively with cycling. Granular composites with one and three reinforcement layers display almost indistinguishable responses. Comparisons across confining pressures further indicate that the influence of initial confinement on damping ratio is negligible.

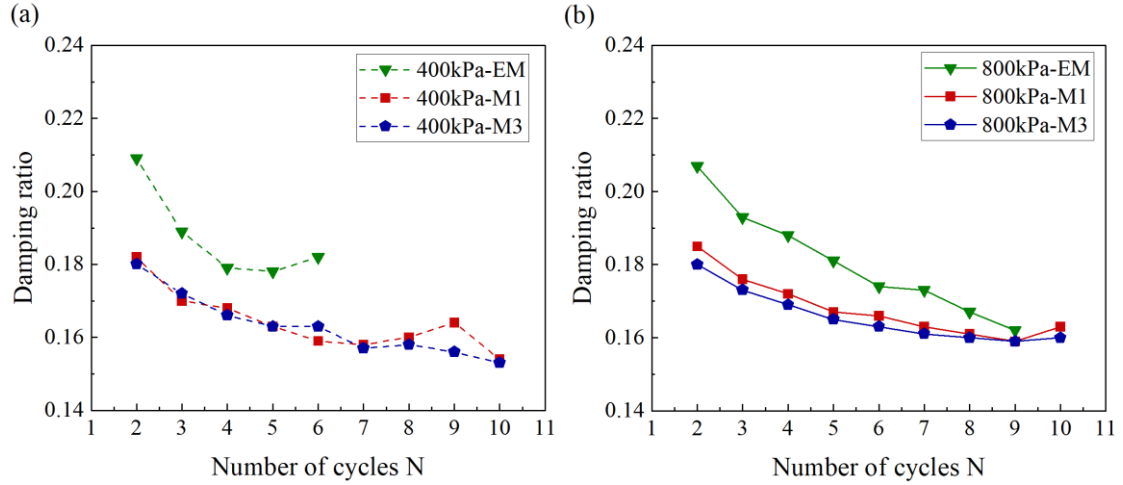


Figure 23. Damping ratio versus number of cycles N with three types of specimens at different initial mean effective pressure under constant-volume condition.

4. Conclusions

A comprehensive program of triaxial test was conducted to evaluate the reinforcing and energy absorption effects of metastructure-embedded granular composites. The study included 48 monotonic constant- σ'_3 and constant-volume tests on specimens with varying initial

states and reinforcement layers, six constant shear stress tests, and 12 strain-controlled cyclic tests under different confining pressures and reinforcement conditions. Stress–strain behavior, effective stress paths, volumetric response, overall energy absorption capacity, mass-normalized specific net energy and damping ratio were systematically examined, together with the residual, phase transformation, and peak states. The principal findings are as follows:

(1) Metastructure confinement, by restraining lateral deformation, substantially modified the stress–strain behavior, volumetric response, and effective stress paths under monotonic loading. Reinforcement led to clear strength enhancement, which increased with the number of layers. In constant-volume tests, stress path shapes were broadly similar and showed no pronounced reduction in mean effective stress, indicating dilation-dominated behavior. Nevertheless, the location of the residual state point varied with reinforcement.

(2) Metastructure inclusion was found to alter the residual state line, with the residual stress ratio M_r increasing as the number of reinforcement layers increased. At the phase transformation state, the stress ratio M_d exhibited a positive correlation with the state parameter and was well captured by an exponential function. In both constant- σ'_3 and constant-volume tests, M_d increased with reinforcement, with a more pronounced trend in the constant-volume case. At the peak state, the stress ratio M_p showed a negative correlation with the state parameter, with data clustering closely around the fitted curve. Overall, M_p increased consistently with reinforcement in both test conditions.

(3) Results of the CSS tests indicated that reinforcement improved the resistance to static instability, with the level of improvement increasing with the number of reinforcement layers.

(4) Under cyclic loading, metastructure enhanced the dynamic response. Reinforced specimens exhibited greater liquefaction resistance, lower excess pore pressure accumulation, higher peak strength, denser hysteresis loops, and broader effective stress path distributions, indicating a reduced susceptibility to instability.

(5) Metastructure inclusion improved the energy absorption capacity of the granular composite across all loading conditions. In constant- σ'_3 monotonic tests, energy absorption was governed primarily by reinforcement layers and confining pressure, with limited influence from initial density. In constant-volume tests, energy absorption also increased with reinforcement, but was strongly affected by initial density and only weakly by confining pressure. In cyclic tests, energy absorption increased with confining pressure, while reinforcement further enhanced capacity under any given pressure. Meanwhile, the damping ratio slightly decreases with metastructure inclusion.

(6) By introducing a mass-normalized specific net energy index w_{net} a highly dynamic efficiency crossover was identified under cyclic loading. Initially, single-layer reinforcement (M1) exhibited a higher per-unit-mass efficiency because the multi-layer (M3) configuration suffered a slight initial reduction caused by the overlapping of their confinement influence zones. However, the robust 3D interlocking network of M3 significantly delayed pore pressure generation and the onset of composite instability. Consequently, M3 sustained active sans-metastructure interaction and continued dissipating energy far beyond the failure point of M1, eventually overtaking it in long-term efficiency.

Appendix

$$\psi = e - e_r$$

$M_d = \frac{q_d}{p'_d}$: q_d , p'_d present the deviator stress and mean effective pressure at phase transformation-state respectively.

Table 3. Test conditions

No.	Legend in Figures	p'_0 (kPa)	e_0	Loading mode	Reinforcement condition	Volume condition
T-1	100kPa, L-EM	100	0.66	Monotonic	Unreinforced	Vented
T-2	200kPa, L-EM	200	0.649	Monotonic	Unreinforced	Vented
T-3	400kPa, L-EM	400	0.638	Monotonic	Unreinforced	Vented
T-4	800kPa, L-EM	800	0.631	Monotonic	Unreinforced	Vented
T-5	100kPa, D-EM	100	0.601	Monotonic	Unreinforced	Vented
T-6	200kPa, D-EM	200	0.589	Monotonic	Unreinforced	Vented
T-7	400kPa, D-EM	400	0.581	Monotonic	Unreinforced	Vented
T-8	800kPa, D-EM	800	0.569	Monotonic	Unreinforced	Vented
T-9	100kPa, L-M1	100	0.661	Monotonic	Single-layer	Vented
T-10	200kPa, L-M1	200	0.651	Monotonic	Single-layer	Vented
T-11	400kPa, L-M1	400	0.637	Monotonic	Single-layer	Vented
T-12	800kPa, L-M1	800	0.623	Monotonic	Single-layer	Vented
T-13	100kPa, D-M1	100	0.6	Monotonic	Single-layer	Vented
T-14	200kPa, D-M1	200	0.592	Monotonic	Single-layer	Vented
T-15	400kPa, D-M1	400	0.577	Monotonic	Single-layer	Vented
T-16	800kPa, D-M1	800	0.571	Monotonic	Single-layer	Vented
T-17	100kPa, L-M3	100	0.663	Monotonic	Triple-layer	Vented
T-18	200kPa, L-M3	200	0.653	Monotonic	Triple-layer	Vented
T-19	400kPa, L-M3	400	0.639	Monotonic	Triple-layer	Vented
T-20	800kPa, L-M3	800	0.626	Monotonic	Triple-layer	Vented
T-21	100kPa, D-M3	100	0.602	Monotonic	Triple-layer	Vented

T-22	200kPa, D-M3	200	0.594	Monotonic	Triple-layer	Vented
T-23	400kPa, D-M3	400	0.582	Monotonic	Triple-layer	Vented
T-24	800kPa, D-M3	800	0.572	Monotonic	Triple-layer	Vented
T-25	100kPa, L-EM	100	0.662	Monotonic	Unreinforced	Isochoric
T-26	200kPa, L-EM	200	0.649	Monotonic	Unreinforced	Isochoric
T-27	400kPa, L-EM	400	0.635	Monotonic	Unreinforced	Isochoric
T-28	800kPa, L-EM	800	0.621	Monotonic	Unreinforced	Isochoric
T-29	100kPa, D-EM	100	0.6	Monotonic	Unreinforced	Isochoric
T-30	200kPa, D-EM	200	0.587	Monotonic	Unreinforced	Isochoric
T-31	400kPa, D-EM	400	0.592	Monotonic	Unreinforced	Isochoric
T-32	800kPa, D-EM	800	0.573	Monotonic	Unreinforced	Isochoric
T-33	100kPa, L-M1	100	0.662	Monotonic	Single-layer	Isochoric
T-34	200kPa, L-M1	200	0.652	Monotonic	Single-layer	Isochoric
T-35	400kPa, L-M1	400	0.638	Monotonic	Single-layer	Isochoric
T-36	800kPa, L-M1	800	0.624	Monotonic	Single-layer	Isochoric
T-37	100kPa, D-M1	100	0.601	Monotonic	Single-layer	Isochoric
T-38	200kPa, D-M1	200	0.592	Monotonic	Single-layer	Isochoric
T-39	400kPa, D-M1	400	0.584	Monotonic	Single-layer	Isochoric
T-40	800kPa, D-M1	800	0.571	Monotonic	Single-layer	Isochoric
T-41	100kPa, L-M3	100	0.662	Monotonic	Triple-layer	Isochoric
T-42	200kPa, L-M3	200	0.655	Monotonic	Triple-layer	Isochoric
T-43	400kPa, L-M3	400	0.641	Monotonic	Triple-layer	Isochoric
T-44	800kPa, L-M3	800	0.625	Monotonic	Triple-layer	Isochoric
T-45	100kPa, D-M3	100	0.602	Monotonic	Triple-layer	Isochoric

T-46	200kPa, D-M3	200	0.592	Monotonic	Triple-layer	Isochoric
T-47	400kPa, D-M3	400	0.585	Monotonic	Triple-layer	Isochoric
T-48	800kPa, D-M3	800	0.575	Monotonic	Triple-layer	Isochoric
T-49	200kPa, EM	200	0.651	CSS	Unreinforced	Vented
T-50	150kPa, EM	150	0.656	CSS	Unreinforced	Vented
T-51	200kPa, M1	200	0.651	CSS	Single-layer	Vented
T-52	150kPa, M1	150	0.656	CSS	Single-layer	Vented
T-53	200kPa, M3	200	0.654	CSS	Triple-layer	Vented
T-54	150kPa, M3	150	0.655	CSS	Triple-layer	Vented
T-55	100kPa, EM	100	0.661	Strain cycles	Unreinforced	Isochoric
T-56	200kPa, EM	200	0.649	Strain cycles	Unreinforced	Isochoric
T-57	400kPa, EM	400	0.641	Strain cycles	Unreinforced	Isochoric
T-58	800kPa, EM	800	0.626	Strain cycles	Unreinforced	Isochoric
T-59	100kPa, M1	100	0.66	Strain cycles	Single-layer	Isochoric
T-60	200kPa, M1	200	0.63	Strain cycles	Single-layer	Isochoric
T-61	400kPa, M1	400	0.633	Strain cycles	Single-layer	Isochoric
T-62	800kPa, M1	800	0.625	Strain cycles	Single-layer	Isochoric
T-63	100kPa, M3	100	0.662	Strain cycles	Triple-layer	Isochoric
T-64	200kPa, M3	200	0.654	Strain cycles	Triple-layer	Isochoric
T-65	400kPa, M3	400	0.636	Strain cycles	Triple-layer	Isochoric
T-66	800kPa, M3	800	0.618	Strain cycles	Triple-layer	Isochoric

651

652 Data Availability Statement

653 All data will be made available on reasonable request.

654

655 **Acknowledgements**

656 The financial support from the Natural Science Foundation of China is appreciated.

657 **Notation**

658 The following symbols are used in this paper:

659 p' = mean effective pressure

660 p_a = standard atmospheric pressure

661 σ'_3 = effective confining pressure

662 Δu = accumulated pore pressure

663 ε_a = shear strain

664 ε_v = volumetric strain

665 $d\varepsilon_a^p$ = incremental plastic shear strain

666 $d\varepsilon_v^p$ = incremental plastic volumetric strain

667 M_r = residual-state ratio

668 M_d = phase transformation-state ratio

669 M_p = peak-state ratio

670 W = cumulative dissipated energy per unit volume

671 w_{net} = mass-normalized specific net energy index of layers

672 m_m = mass of metastructure

673 k = total number of loading increment

674 N_{liq} = cycle number until liquefaction

675 e_r = residual-state void ratio

676 ψ = state parameter

677 A_L = the dissipated energy of a single cycle

678 A_S = ideal elastic strain energy of a single cycle

679

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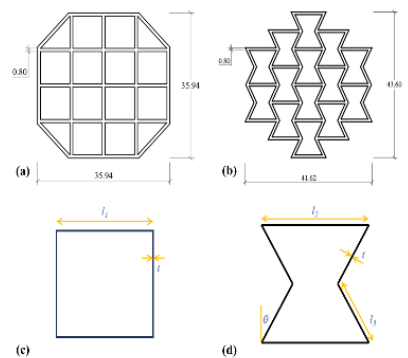
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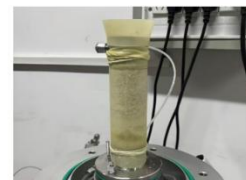
1. Additive manufactured meta-structure with auxetic property
2. Increased deformation resistance of meta-structure granular composites
3. Significant improvement in strength and stability
4. High energy absorbing capability



Pure sand



Composite



Specimen

